

Vanilla LLM Baseline

Overview

Before evaluating retrieval-augmented systems, a direct LLM baseline was established to measure how well a large language model answers biomedical research questions using only its pre-trained knowledge, without access to any external documents or retrieved context. This serves as the lower bound against which all RAG configurations are compared.

Motivation

A vanilla LLM baseline is a necessary reference point in any RAG study. Without it, improvements observed in retrieval-augmented pipelines cannot be attributed confidently to the retrieval component rather than to the generative model itself. If the LLM already performs well on the evaluation set from parametric knowledge alone, retrieval adds relatively little. Conversely, a low baseline score establishes the headroom that retrieval is expected to fill.

Pipeline

The baseline pipeline is straightforward: each question from the evaluation set is passed directly to the language model with a fixed prompt instructing it to answer using its existing knowledge. No documents are retrieved, no context is injected, and the model's response depends entirely on what it has internalised from its training data.

The prompt used was:

Answer the following biomedical research question using your existing knowledge. Provide a precise, evidence-based answer. Do not speculate beyond established scientific knowledge.
Give the answer in 1 or 2 sentences.

The language model used throughout all experiments in this study is **Llama 3.3 70B Versatile**, accessed via the Groq inference API. Groq provides hardware-accelerated inference, making it practical to evaluate a 200-sample dataset within a reasonable time window despite the sequential nature of each individual call.

Evaluation Dataset

The evaluation set consists of 200 questions sampled from the PubMedQA labeled dataset. This golden dataset was constructed once and kept fixed across all experiments, ensuring that every pipeline variant including the vanilla baseline, naive RAG configurations, and query expansion systems sees exactly the same 200 questions. Regenerating the sample mid-experiment would change the question distribution and invalidate cross-experiment comparisons. The golden dataset includes the question text, the expected answer (golden answer), and the reference context passages drawn from the PubMedQA annotations.

Parallelisation Strategy

Evaluating 200 questions sequentially through an external API would be impractically slow and would risk hitting per-key rate limits on a single account. To address this, the evaluation was parallelised using Python's `ThreadPoolExecutor`.

The approach partitions the evaluation set into contiguous slices, one per available API key. Each slice is processed in its own thread, which holds an independent ChatGroq client initialised with the key assigned to that thread. Because each Groq API key carries its own independent daily quota, threads do not contend with one another for rate limit headroom.

`ThreadPoolExecutor` was chosen over Python's `asyncio` for two reasons. First, Jupyter environments already run an active event loop, which means `asyncio.run()` raises a `RuntimeError` if called from within a notebook cell. Second, the ChatGroq client's `invoke()` method is synchronous, so wrapping it in an async coroutine would not yield any concurrency benefit. Since network I/O releases the Python GIL, threads genuinely run concurrently and the throughput gain from multi-key parallelism is real.

In practice, 200 rows were distributed across 10 API keys at 20 rows per key, and all 200 answers were collected successfully in a single parallel run.

Evaluation Metrics

Two reference-based, non-LLM metrics from the RAGAS evaluation framework were used to score the baseline:

BLEU Score measures n-gram overlap between the generated answer and the reference (golden) answer. It is a precision-oriented metric that penalises answers containing n-grams not present in the

reference. For a biomedical QA task where answers are expected to be concise and terminology-specific, BLEU provides a coarse indication of lexical similarity.

ROUGE-L (F-measure) measures the longest common subsequence between the generated and reference answers, normalised by a harmonic mean of precision and recall. It is more sensitive to word ordering than bag-of-n-gram metrics and tends to reward answers that preserve the overall structure of the reference.

For the vanilla LLM, both metrics compare the model's generated answer directly against the golden answer without any retrieved context in the comparison loop.

Additionally, the **AnswerRelevancy** metric from DeepEval was computed using an LLM judge (accessed via the Groq API) to assess whether the generated answers are on-topic and address the question being asked, independent of whether the factual content matches the reference.

Results

Metric	Score
BLEU Score	0.0757
ROUGE-L (F-measure)	0.2113

The BLEU score of 0.0757 indicates very limited lexical overlap between the model's parametric responses and the reference answers. This is expected in a closed-domain biomedical QA setting: the model may produce factually reasonable responses that differ substantially in wording from the PubMedQA expert answers, and BLEU does not account for semantic equivalence.

The ROUGE-L score of 0.2113 is somewhat higher, suggesting that while exact n-gram overlap is low, some structural and phrasal similarity does exist between the model's outputs and the references. This reflects the model's general biomedical literacy from pretraining, which allows it to produce plausible sentence structures and relevant terminology even without grounding in specific retrieved documents.

These scores establish the baseline against which all retrieval-augmented configurations are compared. Any RAG system that outperforms these figures on both metrics is demonstrating a genuine contribution from the retrieval component.